SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	**Assessment:**	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO5 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
	GRAND TOTAL:	20 Marks	

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

(a) Interpret the scenario and identify the type of noise present.
(b) Explain how this degradation affects inspection accuracy.
(c) Recommend an appropriate enhancement technique to remove the interference.
(d) Justify why your selected approach is more suitable than spatial-domain methods.(e) Explain your answer in a logical sequence.

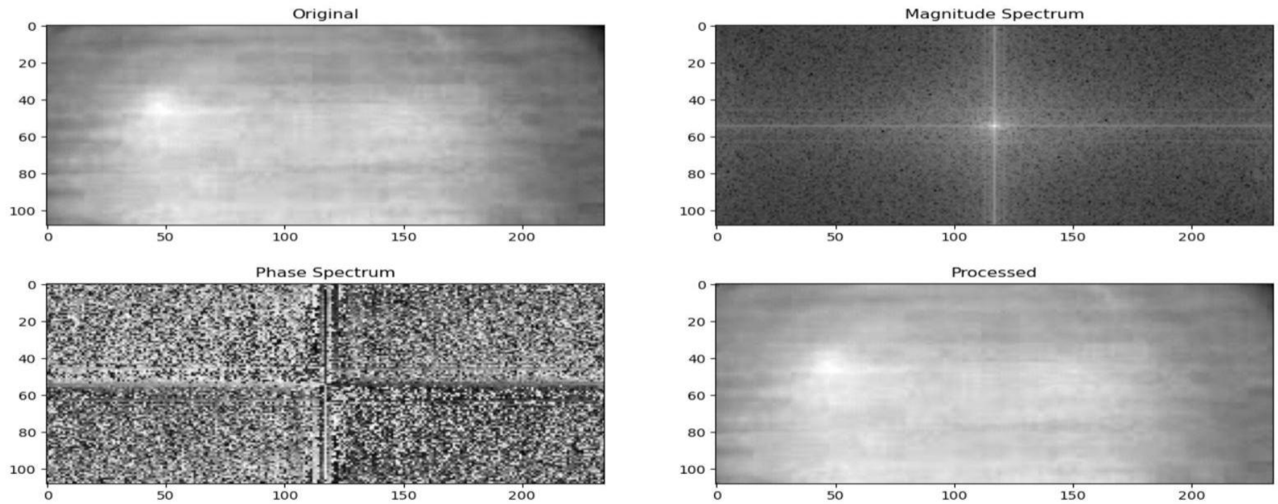
(a) Interpretation & Type of Noise:

In this scenario, an automated inspection system captures images of metal components, but the images contain repetitive horizontal stripes due to electrical interference.

This pattern clearly indicates the presence of Periodic Noise (Structured Noise). Unlike random noise (Gaussian or salt-and-pepper), periodic noise appears as regular repeating patterns, often caused by:

- Electrical interference in sensors
- Power supply fluctuations
- Transmission disturbances

Mathematically, periodic noise can be represented as sinusoidal interference added to the image signal. It is not random, but structured and predictable.



(b) Effect on Inspection Accuracy:

Periodic noise significantly degrades the inspection system:

1. False Defect Detection
   The stripes may resemble cracks or scratches, leading to false positives.

2. Hidden Real Defects
   Actual defects may be masked by the noise pattern.

3. Reduced Edge Clarity
   Important boundaries of components become blurred or distorted.

4. Poor Feature Extraction
   Machine learning models rely on clean features—noise corrupts them.

5. Lower Reliability of Automation
   The system becomes inconsistent, reducing trust in automated inspection.

(c) Recommended Enhancement Technique:

The most effective method is:

Frequency Domain Filtering using Notch Filter

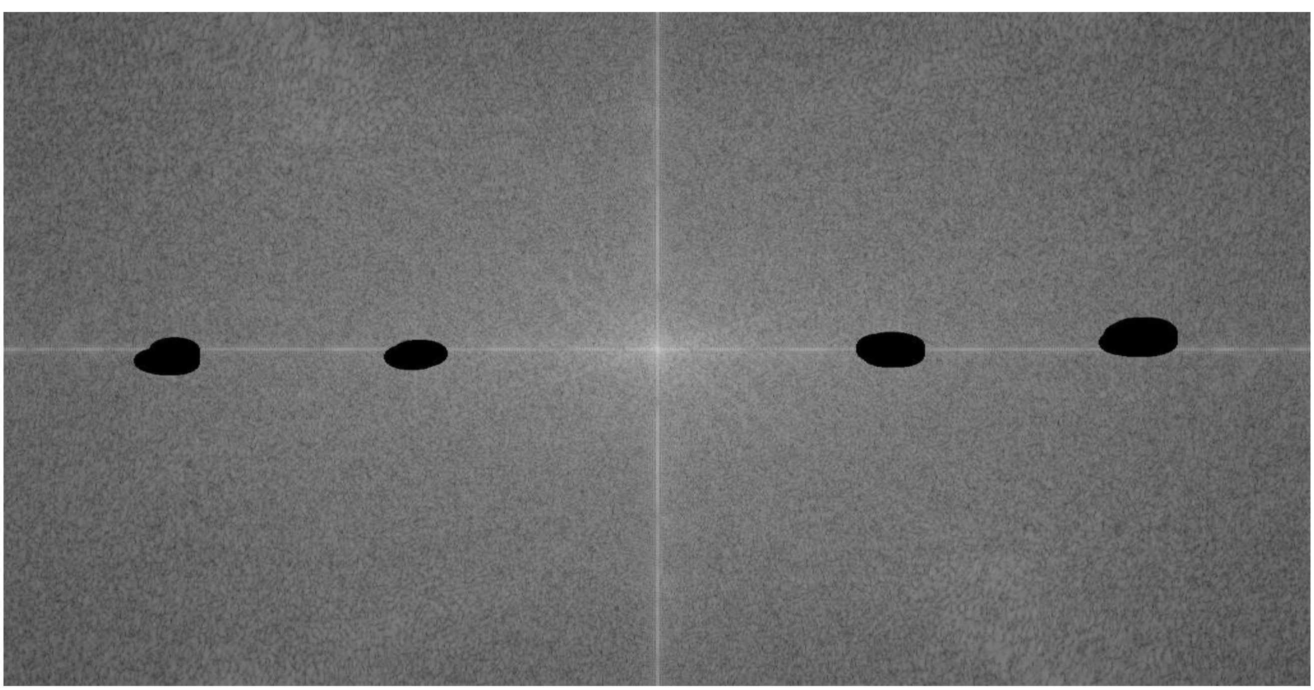
Steps:

1. Convert image to frequency domain using Fourier Transform (FFT)

2. Identify bright spots corresponding to periodic noise

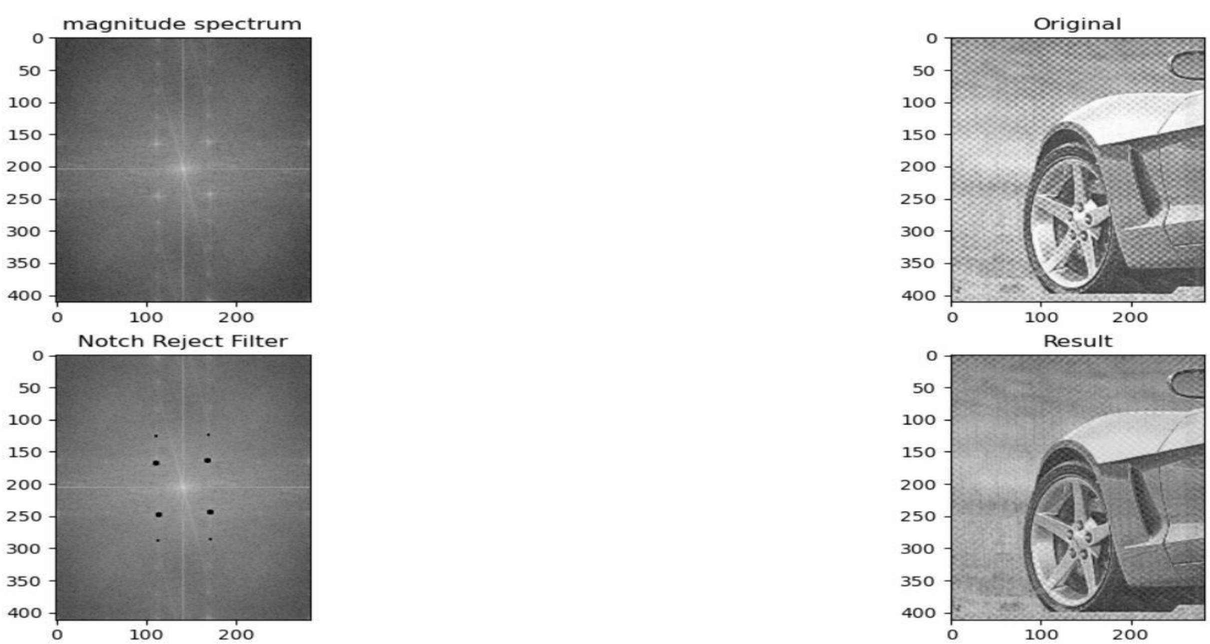
3. Apply Notch Filter to remove those frequencies

4. Convert back using Inverse FFT



(e) Justification: Why Frequency Domain is Better:

Aspect	Frequency Domain (Notch Filter)	Spatial Domain
Noise Removal	Targets exact frequencies	Affects entire image
Precision	High (removes only noise)	Low (blurs image)
Edge Preservation	Maintained	Often degraded
Suitability	Best for periodic noise	Not suitable



Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

(a) Interpret the scenario and explain the challenge involved.

(b) Identify the importance of preserving edges while removing noise.

(c) Recommend a suitable filtering technique.

(d) Justify why your selected filter preserves important image details.

(a) Interpretation & Challenge:

In this scenario, a self-driving car captures road images during heavy rain.

Challenges include:

- Presence of random noise (rain droplets, sensor noise)
- Reduced visibility
- Need to clearly detect lane markings and road edges

The core problem is:

👉 Remove noise while preserving important edges

(b) Importance of Edge Preservation:

Edges are critical for autonomous navigation:

1. Lane Detection
   Lane markings guide vehicle movement.
2. Road Boundary Identification
   Prevents the car from going off-road.
3. Object Detection
   Edges define vehicles, pedestrians, obstacles.
4. Decision Making
   AI models rely on clear structural features.

⚠ If edges are blurred:

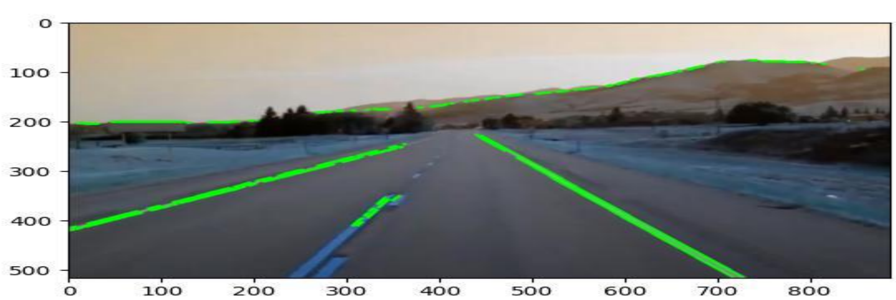
- Lane detection fails
- Navigation becomes unsafe
- Accidents may occur

(c) Recommended Filtering Technique:

Bilateral Filter (Edge-Preserving Filter)

The bilateral filter:

- Smooths noise
- Preserves edges
- Uses both spatial and intensity information



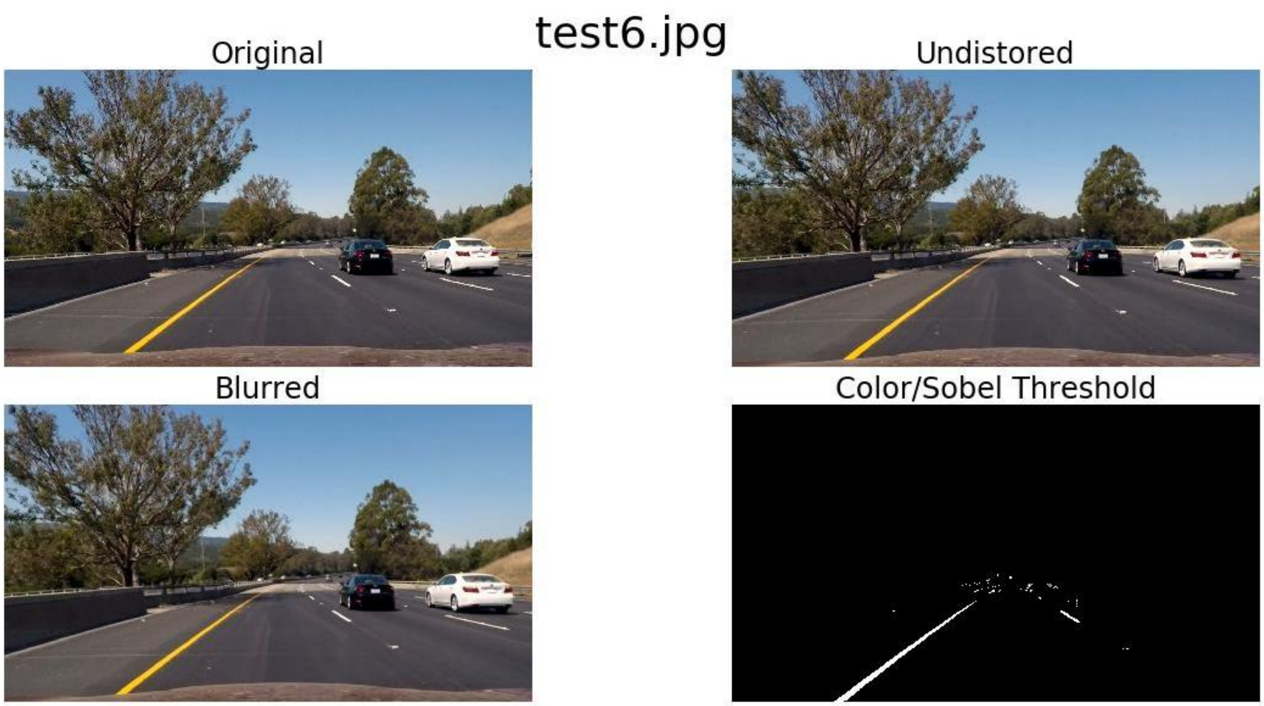
(d) Justification: Why Bilateral Filter Works:

Feature	Bilateral Filter	Gaussian Filter
Noise Removal	Good	Good
Edge Preservation	Excellent	Poor
Blur Effect	Minimal	High
Suitability	Best for navigation	Not suitable

Why it works:

- It considers pixel similarity + distance
- Does not average across edges
- Keeps sharp transitions intact

Thus, lane markings remain clear even after filtering



(e) **Present your explanation clearly and systematically:**

Systematic Explanation:

1. Problem Identification:
   A self-driving car captures road images during heavy rain, introducing random noise and distortions.
2. Nature of Challenge:
   - Rain causes visibility issues and noise
   - Critical features like lane markings and road edges must remain clear
3. Importance of Image Features:
   - Lane markings guide vehicle movement
   - Road edges define safe boundaries
   - These features are essential for navigation decisions
4. Risk of Improper Processing:
   - Standard smoothing filters may blur edges
   - Loss of edge information can lead to navigation errors
5. Recommended Solution:
   Use an edge-preserving filter, specifically the Bilateral Filter.
6. Working Principle:
   - Considers both pixel distance and intensity similarity
   - Smooths noise while avoiding changes across edges
7. Processing Steps:
   - Apply bilateral filtering on the noisy image
   - Reduce rain noise
   - Preserve sharp edges and lane markings
8. Final Outcome:
   - Noise is reduced effectively
   - Edges remain sharp and clear
   - Improves accuracy of lane detection and ensures safe navigation

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	**Course Coordinator**	**Head of Department**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________